

1. SALN Team 1

1.1. Dependency Inventory

1.1.1. Members:

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2. ~~Nonmicroservices Architecture (Laravel's Default MVC)~~

Deprecated Repo: https://github.com/chry607/SALN_Backend

3. Microservices Architecture (Digital Ocean Droplet)

Active Repo: <https://github.com/meezlung/saln-microservices/> (branch: **oop**)

3.1. System Packages (sudo apt install)

Package	Version	Purpose
nginx	1.24.0	Reverse proxy and static file server
php8.4 + extensions	8.4.x	PHP runtime for all three Laravel services
php8.4-cli	8.4.x	PHP CLI (artisan commands)
php8.4-fpm	8.4.x	PHP-FPM process manager
php8.4-mbstring	8.4.x	Multibyte string support
php8.4-xml	8.4.x	XML parsing
php8.4-bcmath	8.4.x	Arbitrary precision math
php8.4-pgsql	8.4.x	PostgreSQL PDO driver
php8.4-curl	8.4.x	HTTP client (Guzzle/curl)
php8.4-zip	8.4.x	ZIP archive support
php8.4-sqlite3	8.4.x	SQLite (local dev/testing)
pdftk	system default	PDF form fill and merge (document service)
postgresql-client	18.1	psql CLI for migrations and DB management
nodejs	v18.19.1	Frontend build toolchain
git	2.43.0	Source control

Note: PHP 8.5.0 was used in practice via the **php.new** / **Herd Lite** musl build (/home/gimislang/.config/herd-lite/bin/php). Composer 2.8.12 was used.

3.2. Auth Service — Composer Dependencies

3.2.1. Production

Package	Constraint	Purpose
laravel/framework	^13.0	Web framework
laravel/sanctum	^4.3	API token authentication (Bearer tokens)
laravel/tinker	^3.0	Artisan REPL
bref/bref	^2.4	PHP Lambda runtime + Serverless plugin
bref/extra-php-extensions	^1.0	Lambda extension layer helper
resend/resend-laravel	^1.3	Resend transactional email driver

3.2.2. Development

Package	Constraint	Purpose
fakerphp/faker	^1.23	Fake data generation for tests/seeders
laravel/pail	^1.2.5	Real-time log tailing
laravel/pint	^1.27	PHP code style fixer
mockery/mockery	^1.6	Test mocking library
nunomaduro/collision	^8.6	CLI error reporting
phpunit/phpunit	^12.5.12	Unit testing framework

3.3. Form Service — Composer Dependencies

3.3.1. Production

Package	Constraint	Purpose
laravel/framework	^13.0	Web framework
laravel/tinker	^3.0	Artisan REPL
bref/bref	^2.4	PHP Lambda runtime + Serverless plugin
bref/extra-php-extensions	^1.0	Lambda extension layer helper

3.3.2. Development

Same as auth service dev dependencies.

3.4. Document Service — Composer Dependencies

3.4.1. Production

Package	Constraint	Purpose
laravel/framework	^13.0	Web framework
laravel/tinker	^3.0	Artisan REPL
brief/bref	^2.4	PHP Lambda runtime + Serverless plugin
barryvdh/laravel-dompdf	^3.1	HTML-to-PDF fallback renderer
mikehaertl/php-pdfkit	^0.14.2	PHP wrapper for pdftk CLI binary
league/flysystem-aws-s3-v3	^3.0	S3 filesystem driver for PDF storage

3.4.2. Development

Same as auth service dev dependencies.

3.5. Frontend — npm Dependencies

3.5.1. Production

Package	Constraint	Purpose
react	^18.3.1	UI library
react-dom	^18.3.1	React DOM renderer
react-router-dom	^7.13.2	Client-side routing
axios	^1.13.6	HTTP client for API calls

3.5.2. Development

Package	Constraint	Purpose
vite	^5.4.1	Build tool and dev server
@vitejs/plugin-react	^4.3.1	React fast-refresh for Vite
eslint	^9.9.0	JavaScript linter
eslint-plugin-react	^7.35.0	React-specific ESLint rules
eslint-plugin-react-hooks	^5.1.0-rc.0	Hooks lint rules
eslint-plugin-react-refresh	^0.4.9	Fast-refresh lint rules
@eslint/js	^9.9.0	ESLint JS config
globals	^15.9.0	Global variable definitions
@types/react	^18.3.3	TypeScript type stubs for React
@types/react-dom	^18.3.0	TypeScript type stubs for React DOM

4. Microservices Architecture (AWS Lambda Function)

Active Repo: <https://github.com/meezlung/saln-microservices/> (branch: **aws-try**)

4.1. Infrastructure Tools

Tool	Version	Purpose
AWS CLI	2.34.50	S3 sync, CloudFront cache invalidation
Serverless Framework	4.36.1	Deploy and manage Lambda functions
Terraform	1.15.4	Provision CloudFront distribution and S3 bucket
Node.js	v18.19.1	Serverless Framework runtime
Composer	2.8.12	PHP dependency install at build time

4.2. Bref / Lambda Runtime Components

Component	Detail
Bref package	bref/bref ^2.4
HTTP runtime	php-84-fpm — handles incoming HTTP events
Console runtime	php-84-console — runs php artisan commands
Base layer ARN	arn:aws:lambda:ap-southeast-2:534081306603:layer:php-84-fpm:40
PostgreSQL layer ARN	arn:aws:lambda:ap-southeast-2:403367587399:layer:pgsql-php-84:4
Custom ini	php/conf.d/pdo_pgsql.ini — loads extension=pdo_pgsql explicitly
OS	provided.al2023 (Amazon Linux 2023)
Lambda timeout (web)	28 s (auth/form), 60 s (document)
Lambda timeout (artisan)	120 s

4.3. AWS Services

Service	Usage
AWS CloudFront	CDN, HTTPS termination, path-based API routing
AWS S3 (saln-frontend-79bf7afb)	Host static React SPA build
AWS S3 (saln-documents-pdfs)	Store generated PDF files
AWS Lambda (saln-auth-service-dev-web)	Auth microservice — php-84-fpm
AWS Lambda (saln-form-service-dev-web)	Form microservice — php-84-fpm
AWS Lambda (collaborator-managed)	Document / PDF generation — JS runtime
Amazon Aurora PostgreSQL Serverless v2	Relational DB cluster in us-east-1 (3 databases)
AWS IAM	Lambda execution roles, S3 write policy for document Lambda

4.4. Composer Dependencies (same packages, Lambda context)

Auth and form service Composer packages are the same as the Digital Ocean section. The key difference is that bref/bref and the Lambda layers are **active** in the AWS deployment, whereas they are installed but unused when running locally with `php artisan serve`.

The document service additionally requires `league/flysystem-aws-s3-v3` for S3 PDF storage — this is only relevant in the Lambda environment since local development uses the local filesystem.

4.5. External Services

Service	Usage
Resend	Transactional email — OTP delivery via <code>onboarding@upcsweb.dev</code> (set up by Prof. Sir Rom Feria)
<code>sa1n1a.upcsweb.dev</code>	CNAME alias for CloudFront distribution (DNS managed by Prof. Sir Rom Feria)
<code>sa1n1.upcsweb.dev</code>	Digital Ocean droplet domain; also used as fallback API origin in CloudFront